

z/OS Subsystems and JES

Primary and secondary subsystems, job control

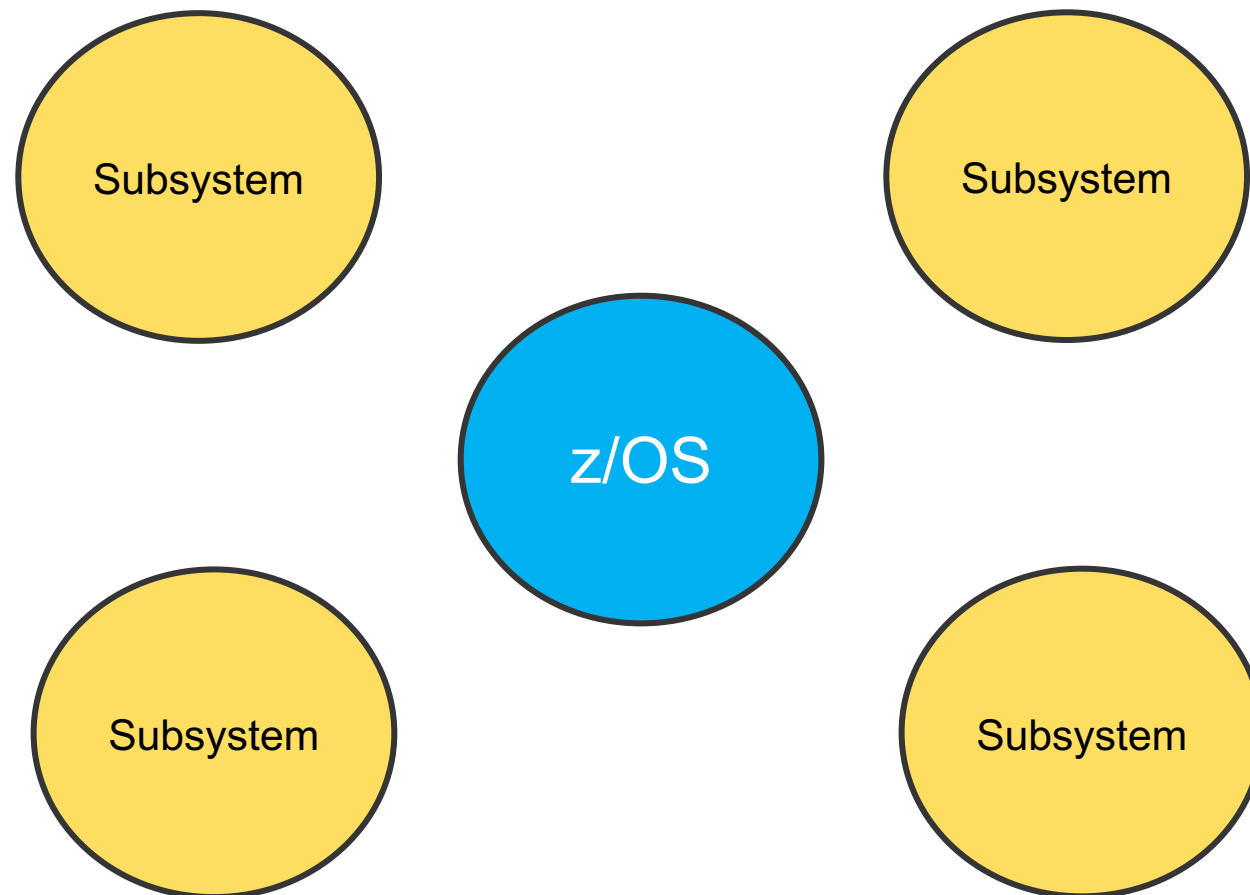




Z/OS SUBSYSTEMS

Like nearly all operating systems that support general-purpose computing, z/OS consists of core functionality that handles basic operations and otherwise depends on services that are called by the core functionality. Also like nearly all operating systems, z/OS can run an application as a service.

On z/OS, services are called *subsystems*.

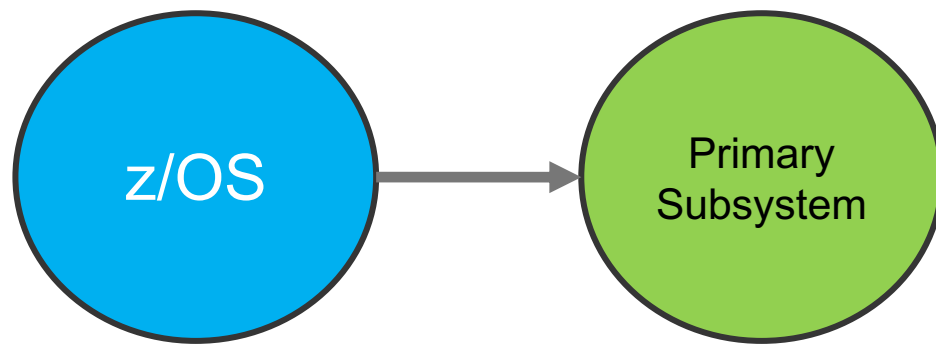


Z/OS SUBSYSTEMS

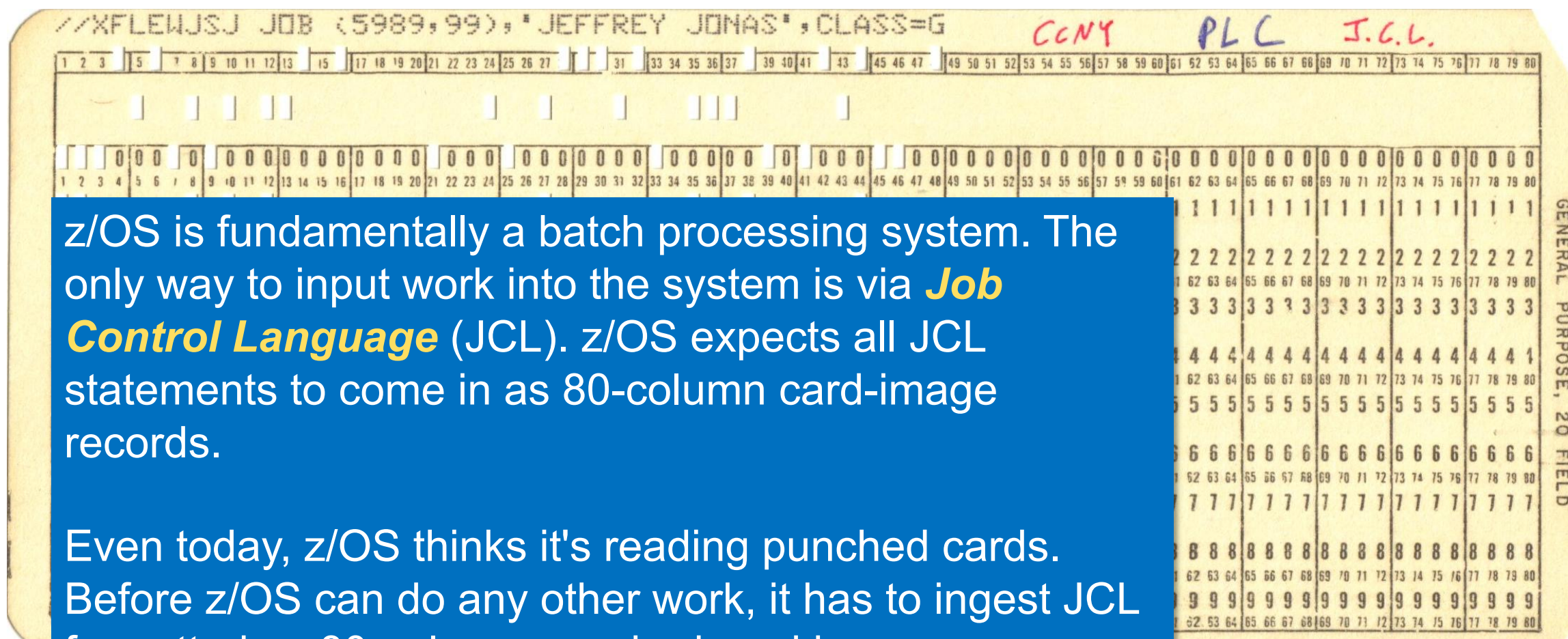
Most operating systems initiate a select list of critical services on startup so that the system will be usable. Some services may ship with the operating system and are considered fundamental to the system. Others may be third-party add-ons. We can usually substitute a different service for one that is provided by default.

One service is special – it's the first one to be initiated during operating system startup, and it may be responsible for starting the other services that must be resident before the system is usable.

On z/OS, the main service that is started during system initialization is called the *primary subsystem*.



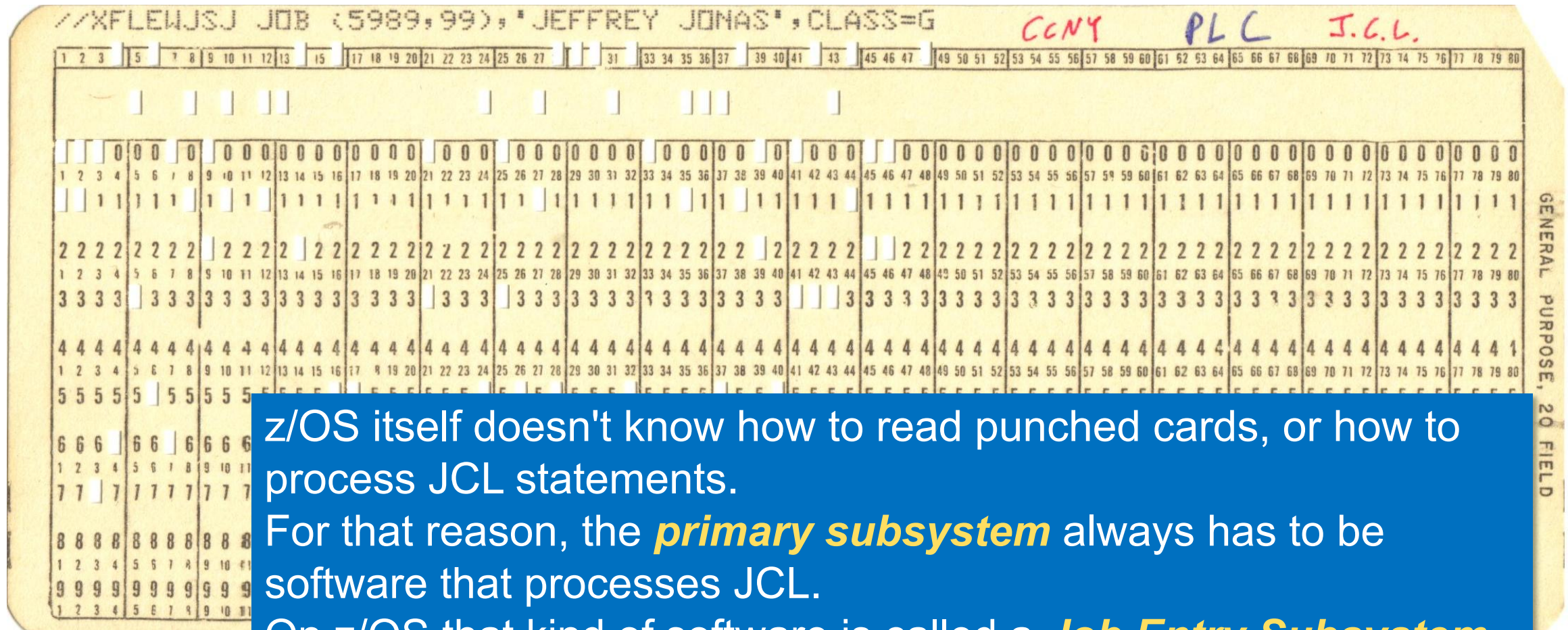
Z/OS JOB CARD (SAMPLE)



z/OS is fundamentally a batch processing system. The only way to input work into the system is via **Job Control Language** (JCL). z/OS expects all JCL statements to come in as 80-column card-image records.

Even today, z/OS thinks it's reading punched cards. Before z/OS can do any other work, it has to ingest JCL formatted as 80-column punched card images.

Z/OS JOB CARD (SAMPLE)

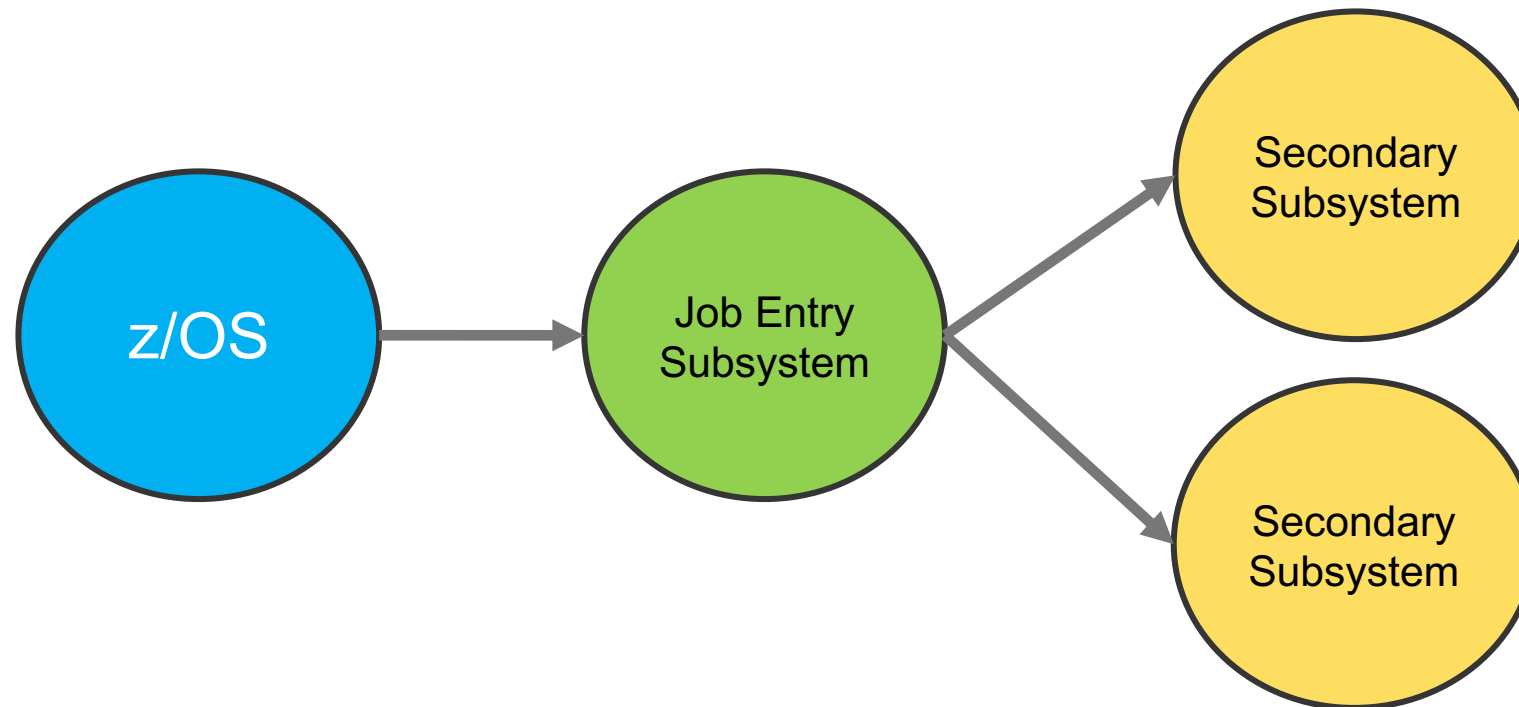


Z/OS SUBSYSTEMS

On most operating systems, the first service (typically running as PID 1 and having no parent process) starts the other services that are necessary to bring the system up to a "target" operating state.

z/OS is conceptually similar, but the implementation is a little different. Secondary subsystems are started by running jobs. To run a job you need Job Control Language (JCL). To process JCL you need a Job Entry Subsystem (JES). So, on z/OS, the primary subsystem is always a Job Entry Subsystem.

On z/OS, *secondary subsystems* are started by running jobs found in a predefined location on the system.



JES2 AND JES3

Some facts about JES2 and JES3:

- There are two flavors of Job Entry Subsystem – JES2 and JES3.
- They are not version 2 and version 3 of the same software.
- JES2 is designed for large installations that run multiple instances of z/OS.
- JES3 is designed for smaller installations that run a single instance of z/OS.
- While Job Control Language (JCL) controls jobs, Job Entry Control Language (JECL) controls the Job Entry Subsystem.
- JES2 and JES3 have different JECL.
- Today, it's rare to find an installation that uses JES3. The surviving mainframe installations are the largest ones, that have no practical or cost-effective alternative to running mainframe systems. They use JES2.

REFERENCES

Here are some links to documentation.

z/OS Initialization Process

<https://www.ibm.com/docs/en/zos/3.1.0?topic=overview-initialization-process>

z/OS: Job Entry Subsystem

<https://www.ibm.com/docs/en/zos/3.1.0?topic=sets-job-entry-subsystem>

WHAT YOU KNOW AND WHAT YOU CAN DO

- You know z/OS is fundamentally a batch processing system with support for interactive applications added on.
- You know the Job Entry Subsystem (JES) is always the primary subsystem on z/OS.
- You know secondary subsystems are started *via* Job Control Language (JCL), and that is why JES must be running first.
- You know JES2 is for larger installations running multiple z/OS instances.

WHAT'S NEXT

- Job Control Language (JCL).